Figure report

Generated 09-Sep-2026 17:54:31 on R5611351, MATLAB 26.1.0.3312084 (R2026a) Update 4.

Figure root	`C:\Users\m218089\Desktop\github_repos\FractionalReservoir`
Sections	20
Images	26
commit_short	`e01361c`
branch	`refactorRunAll`
dirty	`false`
Run directory	`C:\Users\m218089\Desktop\github_repos\FractionalReservoir`
Preset	`celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_`
Run mode	`fast`
Result	20/20 succeeded, 13/13 in-paper
Elapsed	5.4 min

Values above are echoed from manifest.md, which is the record of the run.

Contents

doc_tables

Inlined from `doc_tables/adaptation_conditions.md`.

Adaptation conditions

Preset: `celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25` · **Model class:** `SRNNCellTypePairs`

> GENERATED by `write_manuscript_tables` from a model BUILT from that preset under the `sfa_and_std` condition. Do not edit by hand – every value here is read off the object, so it cannot disagree with what the sweeps ran. Regenerate after changing the preset.

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Every sweep runs each grid point under all 3 regimes, on the **same network** (the weight seed is shared), so the comparison is paired.

Condition	SFA timescales τ_a (s)	Depressing routes	Facilitating routes
No Adaptation	none	none	none
Single-Timescale Adaptation	0.25	EE, EI, IE, II	none
Multiple-Timescale Adaptation	0.25, 1.581, 10	EE, EI, IE, II	none

Inlined from `doc_tables/adaptation_conditions_table.md`.

Condition	SFA timescales τ_a (s)	Depressing routes	Facilitating routes
No Adaptation	none	none	none
Single-Timescale Adaptation	0.25	EE, EI, IE, II	none
Multiple-Timescale Adaptation	0.25, 1.581, 10	EE, EI, IE, II	none

Inlined from `doc_tables/equation_table.md`.

SRNN model equations

Preset: `celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25` · **Model class:** `SRNNCellTypePairs`

> GENERATED by `write_manuscript_tables` from a model BUILT from that preset under the `sfa_and_std` condition. Do not edit by hand – every value here is read off the object, so it cannot disagree with what the sweeps ran. Regenerate after changing the preset.

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System equations

$$\mathrm{d}x_i = \frac{-x_i + u_i + \sum_j w_{ij}\, r_j \prod_{m=1}^{2} b_{jm}}{\tau_d}\,\mathrm{d}t + \frac{\sigma_u}{\tau_d}\,\mathrm{d}W_i$$

Additive Wiener noise enters **only** x, which keeps the diffusion constant: Ito and Stratonovich coincide, the Milstein term vanishes, the variational equation is untouched, and the noise cancels in Benettin's trajectory difference. σ_u is input-referred, so it is comparable to the stimulus amplitude.

$$r_i = \phi\left(x_i - a_{0_i} - \frac{c}{K}\sum_{k=1}^{K} a_{ik}\right)$$

ϕ is centred on zero; a_{0_i} is the **setpoint** (`S_c`, 0.2 here), per neuron when `mu_S_c` / `sigma_S_c` are set.

The rate r_i is the **pre-depression** output of the nonlinearity. Depression and facilitation enter presynaptically, as factors multiplying r_j inside the recurrent sum, so both mechanisms are driven by the raw rate r_i and not by the depressed output.

$$\frac{\mathrm{d}a_{ik}}{\mathrm{d}t} = \frac{-a_{ik} + r_i}{\tau_{a_k}}$$

$$\frac{\mathrm{d}b_{im}}{\mathrm{d}t} = \frac{1 - b_{im}}{\tau_{rec_m}} - \frac{b_{im}\, r_i}{\tau_{rel_m}}, \qquad m = 1, \ldots, 2$$

$$\theta_j = r_j \prod_{m=1}^{2} b_{jm}$$

θ_j is the **synaptic output**: the presynaptic rate after depression, and the quantity the recurrent sum above transmits.

The depression timescales enter as a **product**, which is the reason for using more than one: 2 of them SQUARE the depression at steady state, so the extra timescale buys DEPTH and not merely a slower approach. There is no split of τ that would make them match a single timescale.

With $\tau_{rec}/\tau_{rel} = 8$ on every timescale, each factor relaxes to $b_k(r) = 1/(1 + 8\,r)$, so at $r = 1$ a single timescale gives a gain of 0.1111 (a 9-fold reduction against an undepressed synapse) while 2 give 0.01235 – a **81-fold** reduction, exactly the square.

Adaptation, by contrast, enters as a **sum**, and is normalised by its timescale count: the rate subtracts $(c/K)\sum_{k=1}^{K} a_k$. Since every a_k relaxes to the rate, $\sum_k a_k \to Kr$ regardless of the τ_k, so the steady state is $c\,r$ whatever K is – adding SFA timescales changes the route, not the destination. c is therefore the **total** adaptation budget. No such normalisation applies to depression, which multiplies rather than sums.

The sparse weights are drawn element-wise and scaled by the synaptic gain g:

$$w_{ij} = g\, S_{ij}\left(\tilde{\sigma}_{ij}\,\xi_{ij} + \tilde{\mu}_{ij}\right), \qquad \xi_{ij} \sim \mathcal{N}(0,1), \qquad S_{ij} \sim \mathrm{Bernoulli}(\alpha)$$

with $(\tilde{\mu}, \tilde{\sigma})$ indexed by **(postsynaptic, presynaptic)** cell type, and given as multiples of $F = 1/\sqrt{n\,\alpha(2-\alpha)}$.

Parameters as run

Symbol	Name	Value	Units
n	Network size	500	neurons
α	Connection probability	0.2	–
indegree	Expected in-degree	100	synapses
f	Fraction per cell type	`[0.5 0.5]`	–
τ_d	Dendritic time constant	0.1	s
g	Synaptic gain	1	–
$\tilde{\mu}$	Weight means (post, pre)	`[8.25 -8.25;8.25 -8.25]`	multiples of F
$\tilde{\sigma}$	Weight std devs (post, pre)	`[1.5 1.5;1.5 1.5]`	multiples of F
ϕ	Nonlinearity	`piecewise`	–
S_c	Nonlinearity setpoint	0.2	–
S_a	Piecewise slope parameter	0.8	–
K	SFA timescales per type	`[3 3]`	–
τ_a (E)	SFA time constants	`[0.25 1.581 10]`	s
τ_a (I)	SFA time constants	`[0.25 1.581 10]`	s
c	SFA coupling per type	`[0.5 0.5]`	–
τ_{rec} (E to E)	STD recovery	`[2 4]`	s
τ_{rel} (E to E)	STD release	`[0.25 0.5]`	s
τ_{rec} (E to I)	STD recovery	`[2 4]`	s
τ_{rel} (E to I)	STD release	`[0.25 0.5]`	s
τ_{rec} (I to E)	STD recovery	`[2 4]`	s
τ_{rel} (I to E)	STD release	`[0.25 0.5]`	s
τ_{rec} (I to I)	STD recovery	`[2 4]`	s
τ_{rel} (I to I)	STD release	`[0.25 0.5]`	s
σ_u	Input-referred noise	0.025	input units
σ_x	Stationary std of x	0.0559017	–
f_s	Sample rate	400	Hz

Inlined from `doc_tables/parameters_table.md`.

Symbol	Name	Value	Units
n	Network size	500	neurons
α	Connection probability	0.2	–
indegree	Expected in-degree	100	synapses
f	Fraction per cell type	`[0.5 0.5]`	–
τ_d	Dendritic time constant	0.1	s
g	Synaptic gain	1	–
$\tilde{\mu}$	Weight means (post, pre)	`[8.25 -8.25;8.25 -8.25]`	multiples of F
$\tilde{\sigma}$	Weight std devs (post, pre)	`[1.5 1.5;1.5 1.5]`	multiples of F
ϕ	Nonlinearity	`piecewise`	–

Symbol	Name	Value	Units
S_c	Nonlinearity setpoint	0.2	–
S_a	Piecewise slope parameter	0.8	–
K	SFA timescales per type	`[3 3]`	–
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c	SFA coupling per type	`[0.5 0.5]`	–
τ_{rec} (E to E)	STD recovery	`[2 4]`	s
τ_{rel} (E to E)	STD release	`[0.25 0.5]`	s
τ_{rec} (E to I)	STD recovery	`[2 4]`	s
τ_{rel} (E to I)	STD release	`[0.25 0.5]`	s
τ_{rec} (I to E)	STD recovery	`[2 4]`	s
τ_{rel} (I to E)	STD release	`[0.25 0.5]`	s
τ_{rec} (I to I)	STD recovery	`[2 4]`	s
τ_{rel} (I to I)	STD release	`[0.25 0.5]`	s
σ_u	Input-referred noise	0.025	input units
σ_x	Stationary std of x	0.0559017	–
f_s	Sample rate	400	Hz

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`fig_adaptation_methods/Fig_single_neuron_SFA_STD.png`

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`fig_energy_landscape/panelA_energy_landscape.png`

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`fig_example_timeseries/fig_example_timeseries.png`

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`fig_FI_curve/Fig_FI_curve.png`

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`fig_introductory_concepts/statetraces/panelA_bottom_traces.png`

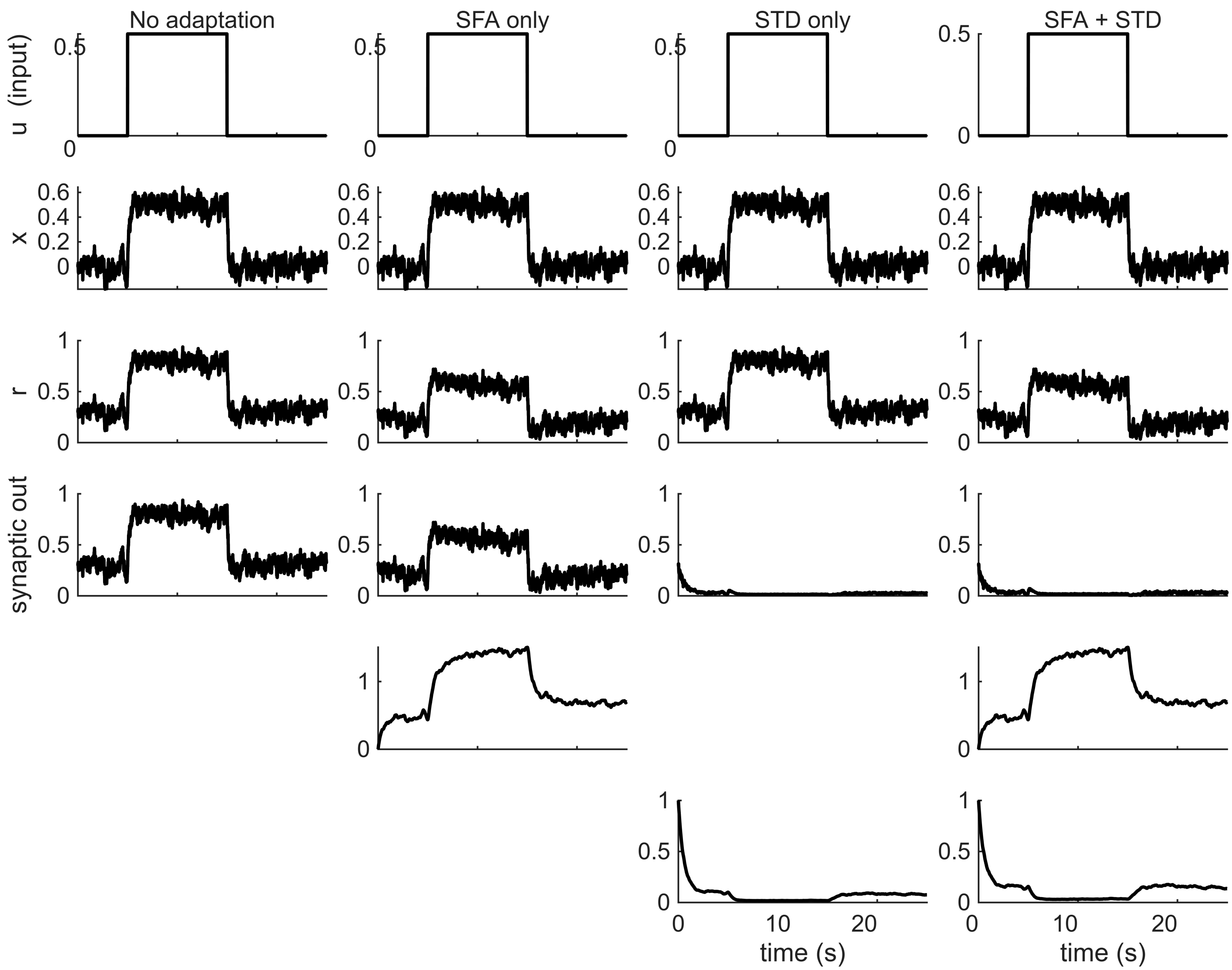
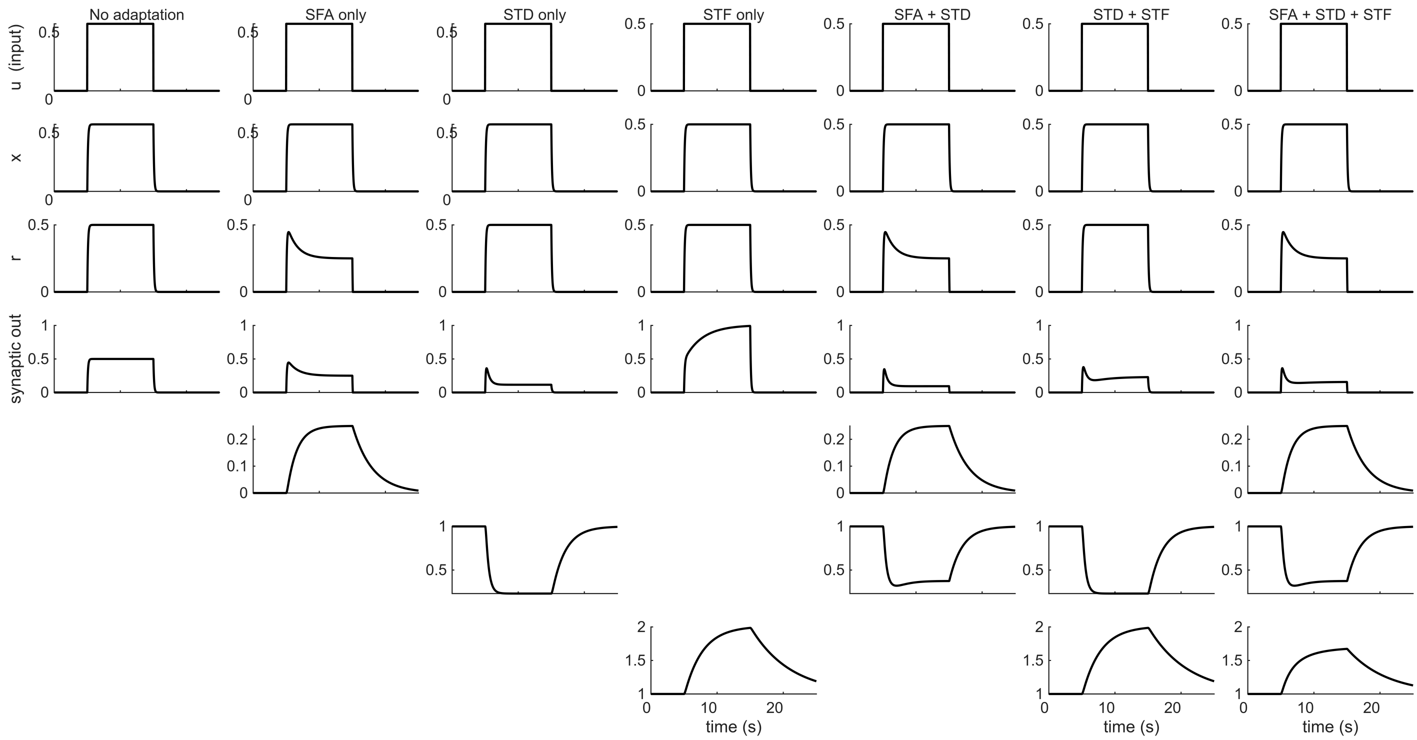


Figure 1: fig_adaptation_methods/Fig_single_neuron_SFA_STD.png



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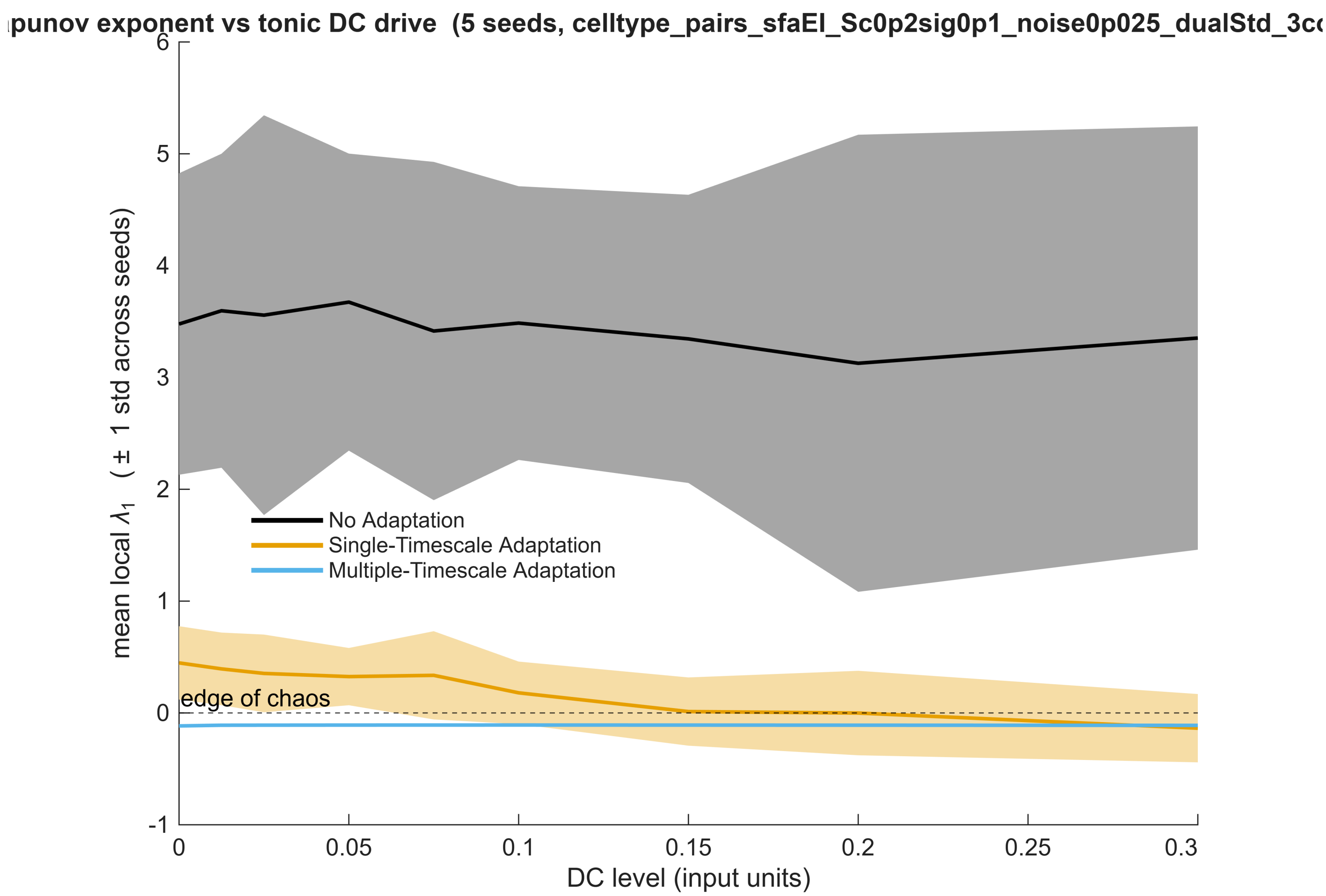


Figure 3: fig_dc_lle/fig_dc_lle.png

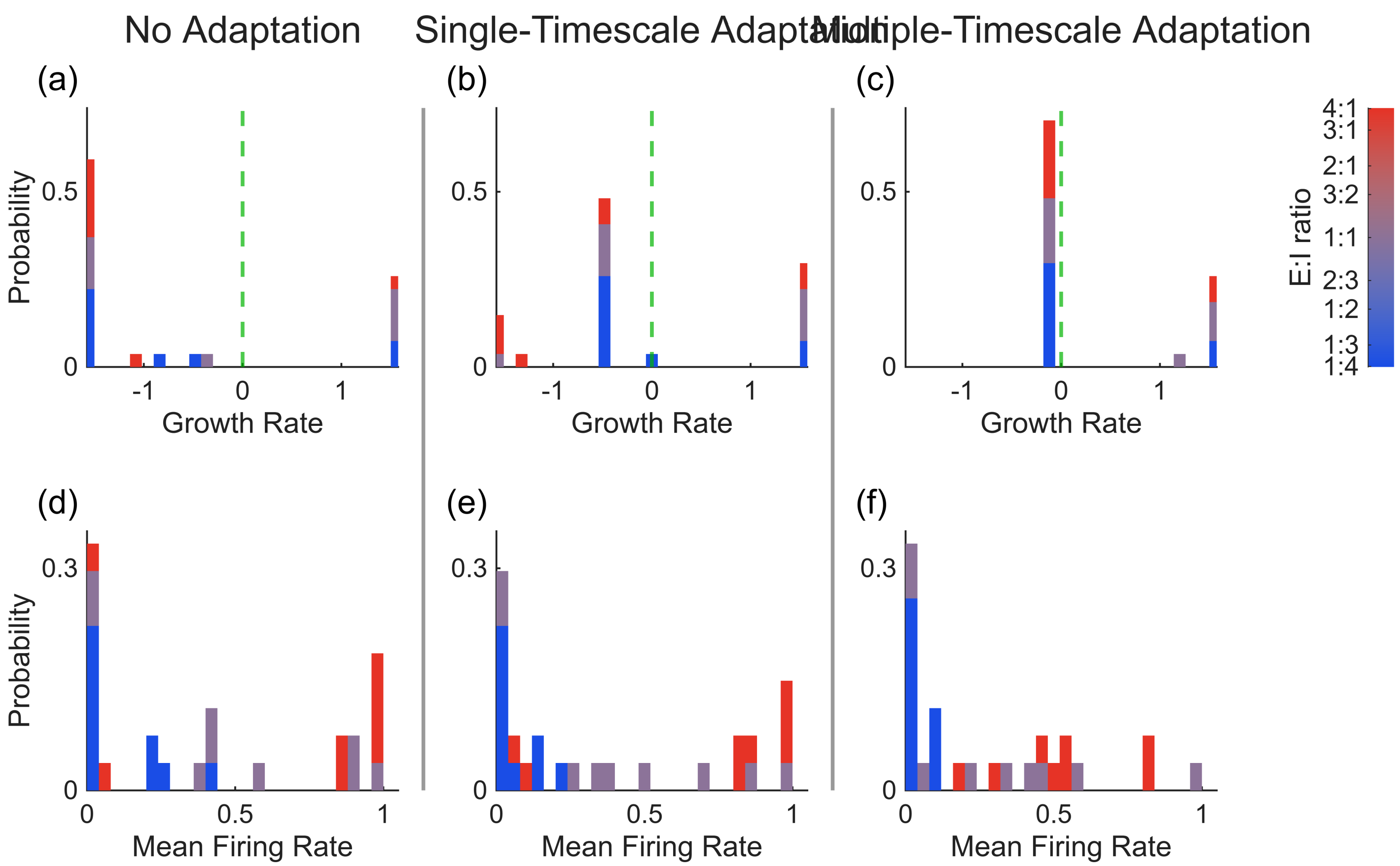


Figure 4: fig_EI_param_space/Fig_EI_ParamSpace.png

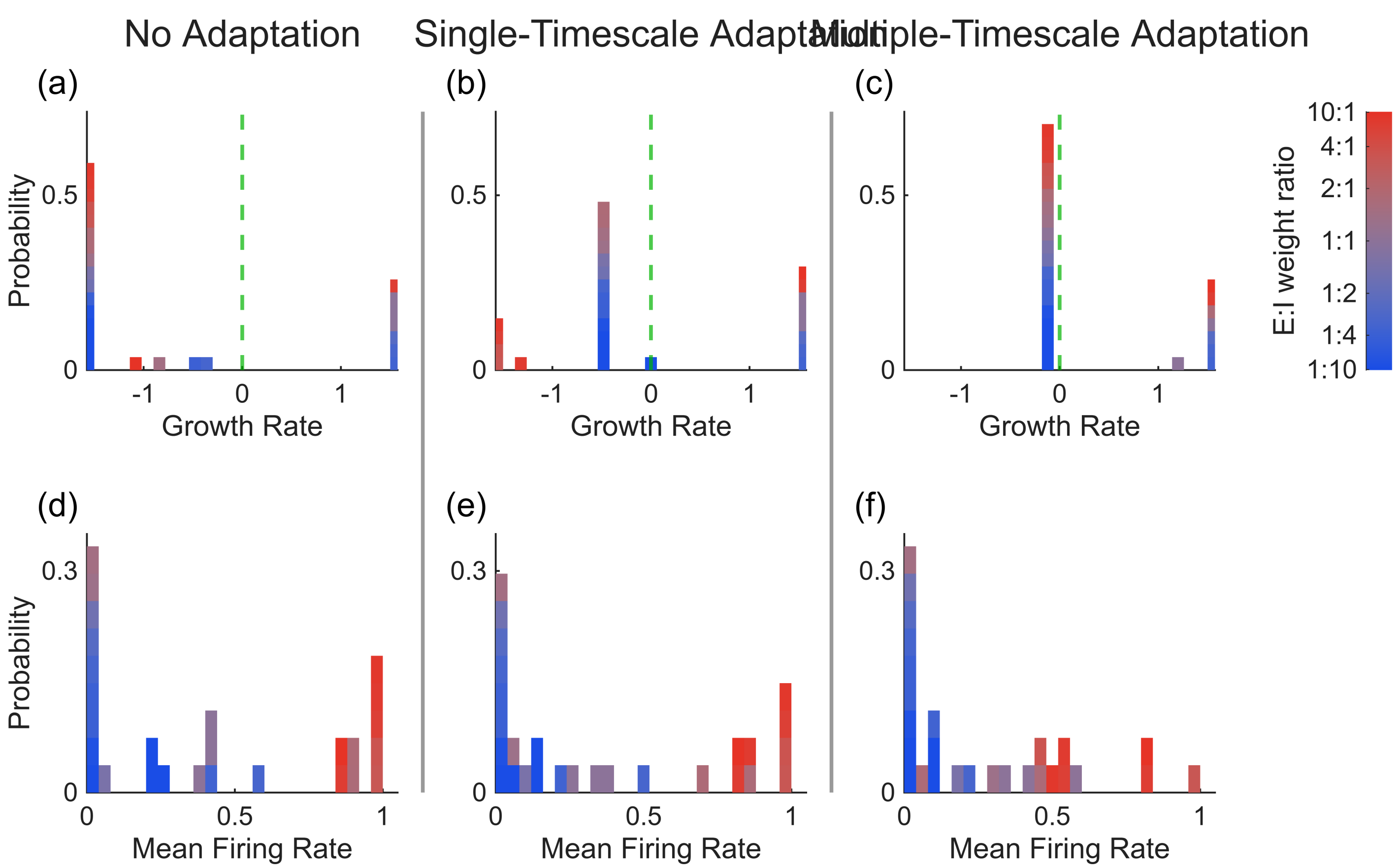


Figure 5: fig_EI_weights_param_space/Fig_EI_Weights_ParamSpace.png

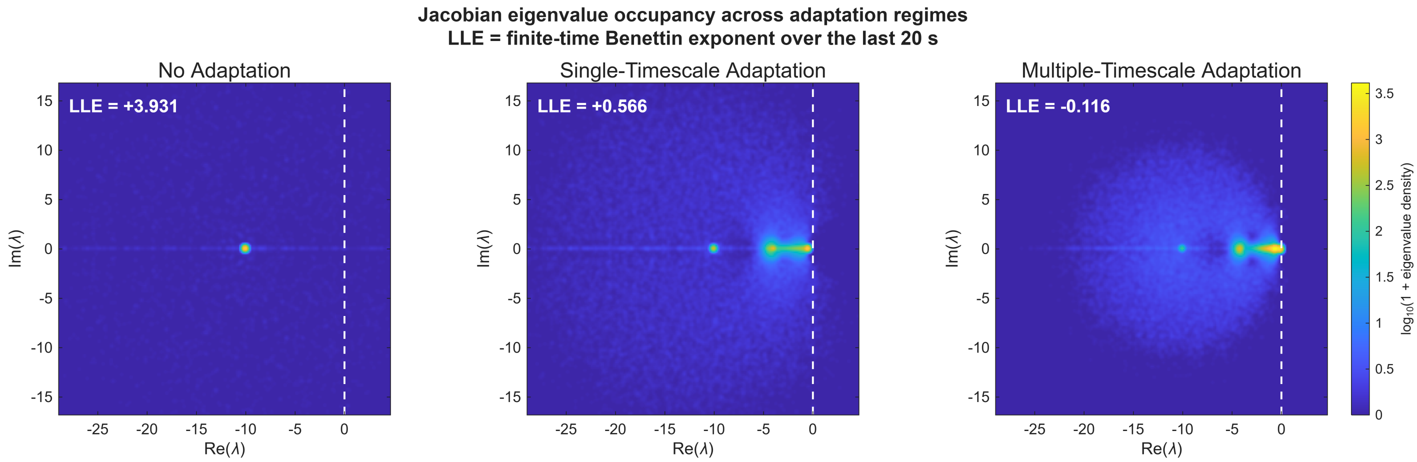


Figure 6: fig_eig_heatmap/fig_eig_heatmap.png

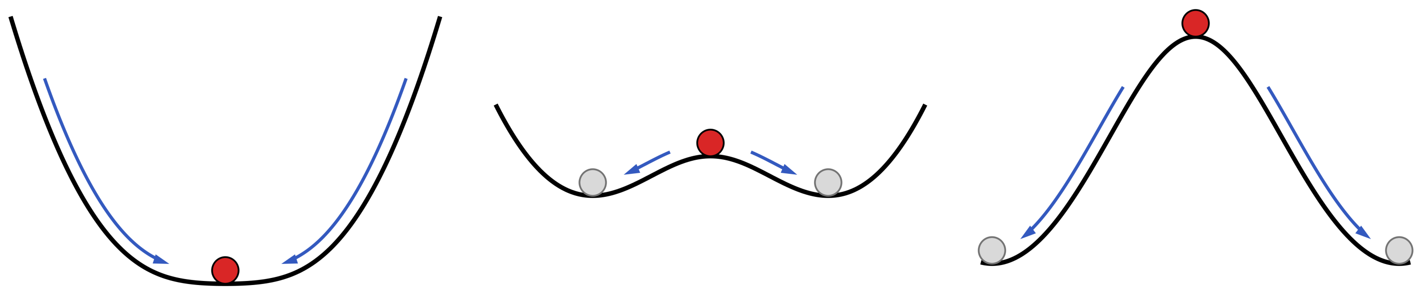
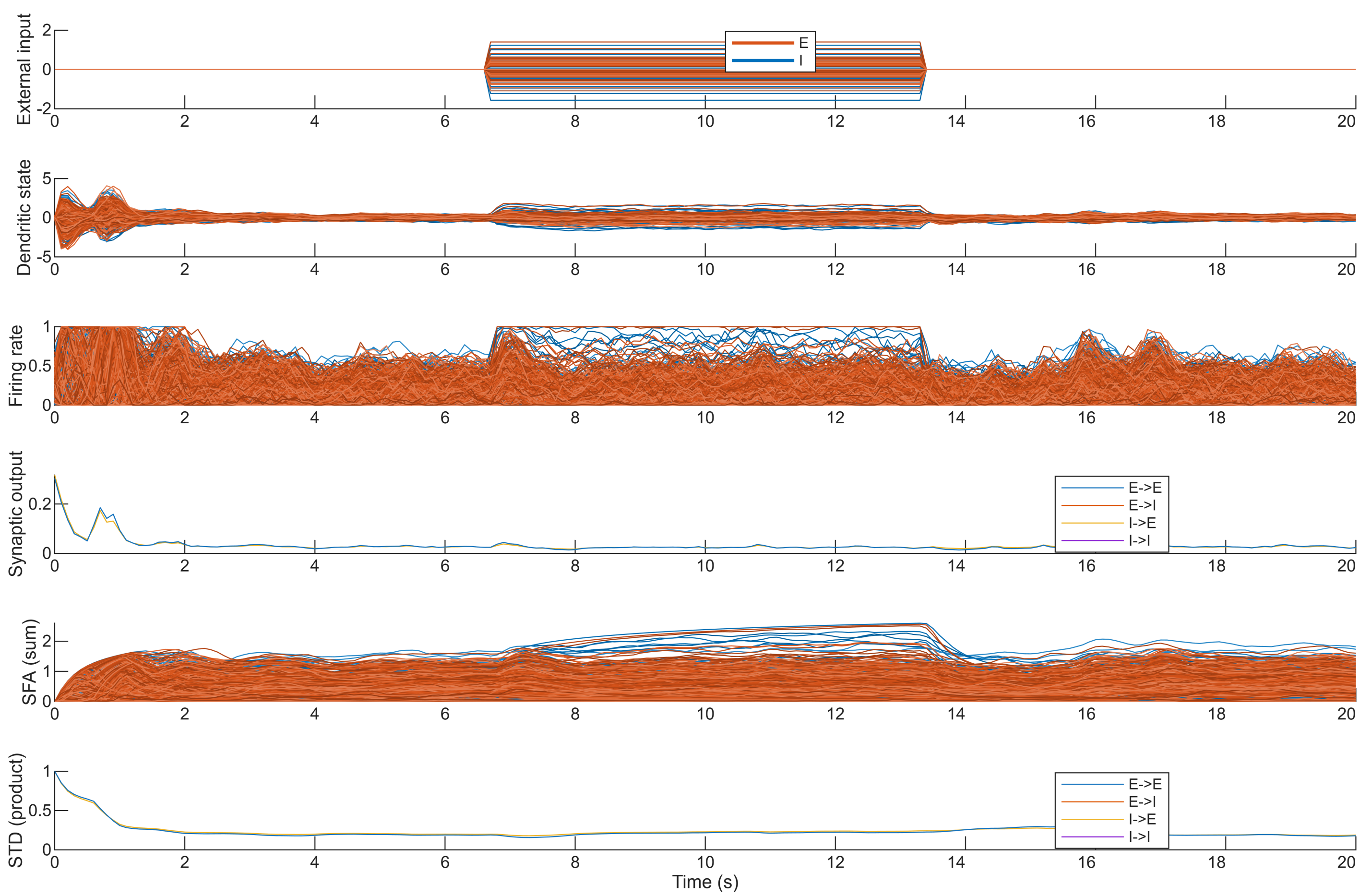


Figure 7: fig_energy_landscape/panelA_energy_landscape.png



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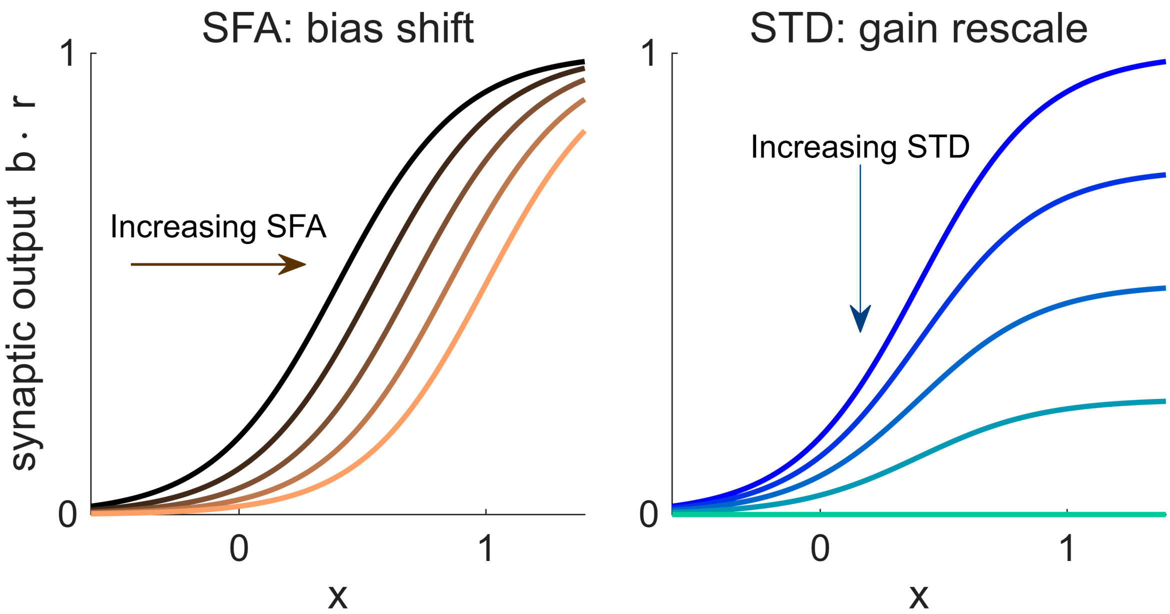

Figure 9: fig_FI_curve/Fig_FI_curve.png

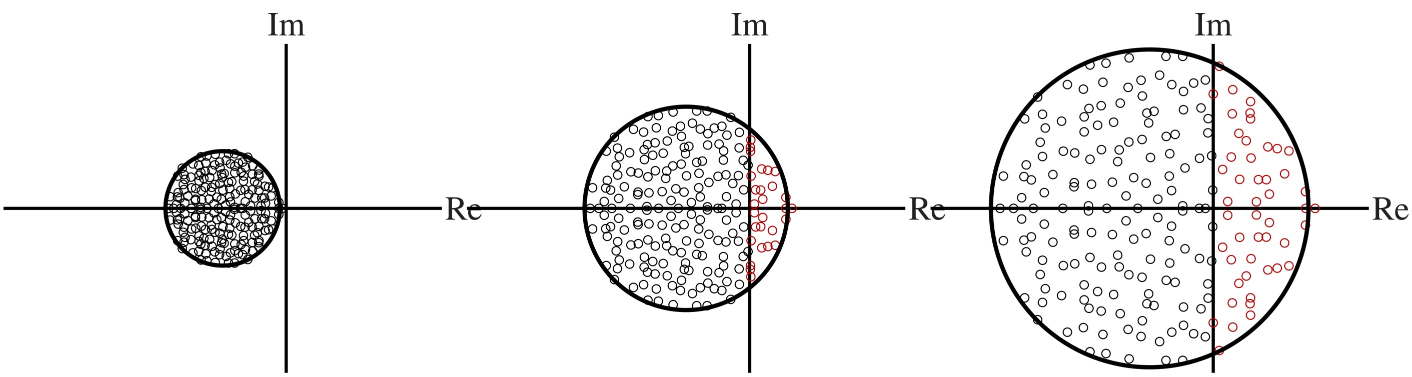

Figure 10: fig_introductory_concepts/eigenspectra/panelA_eigenspectrum.png

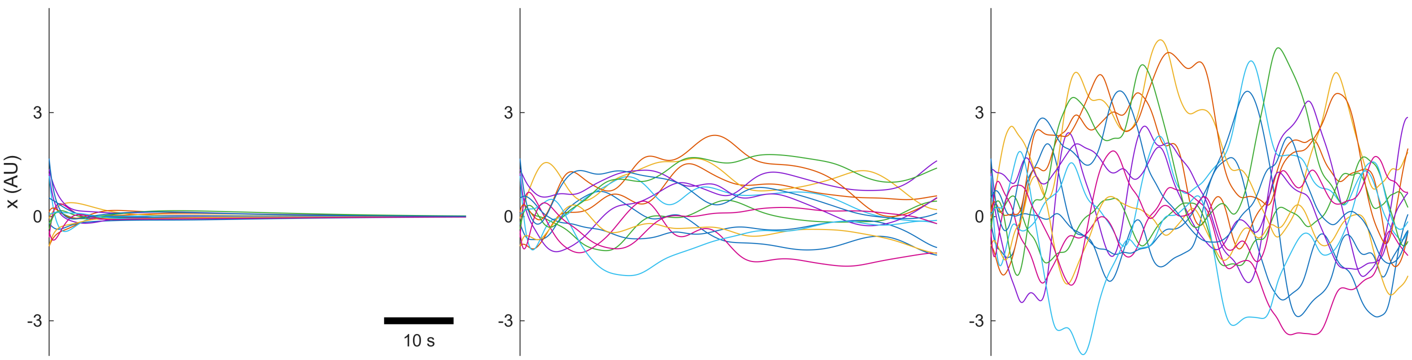

Figure 11: fig_introductory_concepts/statetraces/panelA_bottom_traces.png

fig__memory__capacity

`fig_memory_capacity/Fig_Memory_Capacity.png`

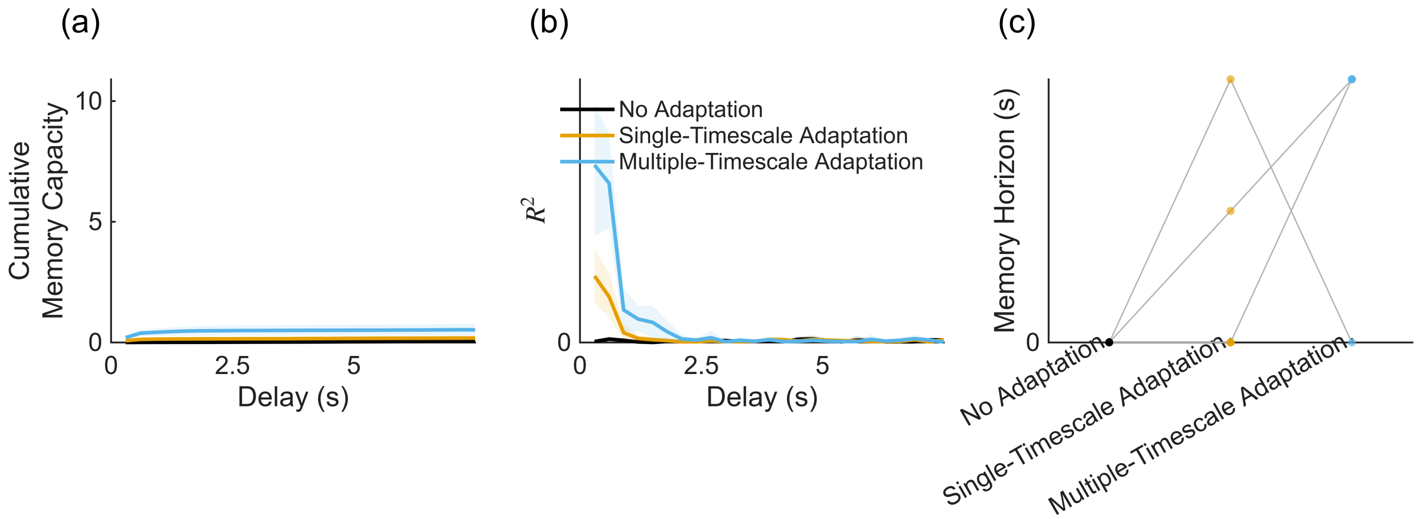

Figure 12: fig__memory__capacity/Fig__Memory__Capacity.png

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`fig_memory_capacity_example/Fig_MC_Example.png`

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`fig_param_space_allStd/Fig_ParamSpace_allStd.png`

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`fig_sensitivity_analysis_allStd/Fig_Sensitivity_LLE_mu.png`

`fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_core.png`

`fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_mu.png`

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`fig_sfa_EOC_allStd/Fig_SFA_EOC_allStd.png`

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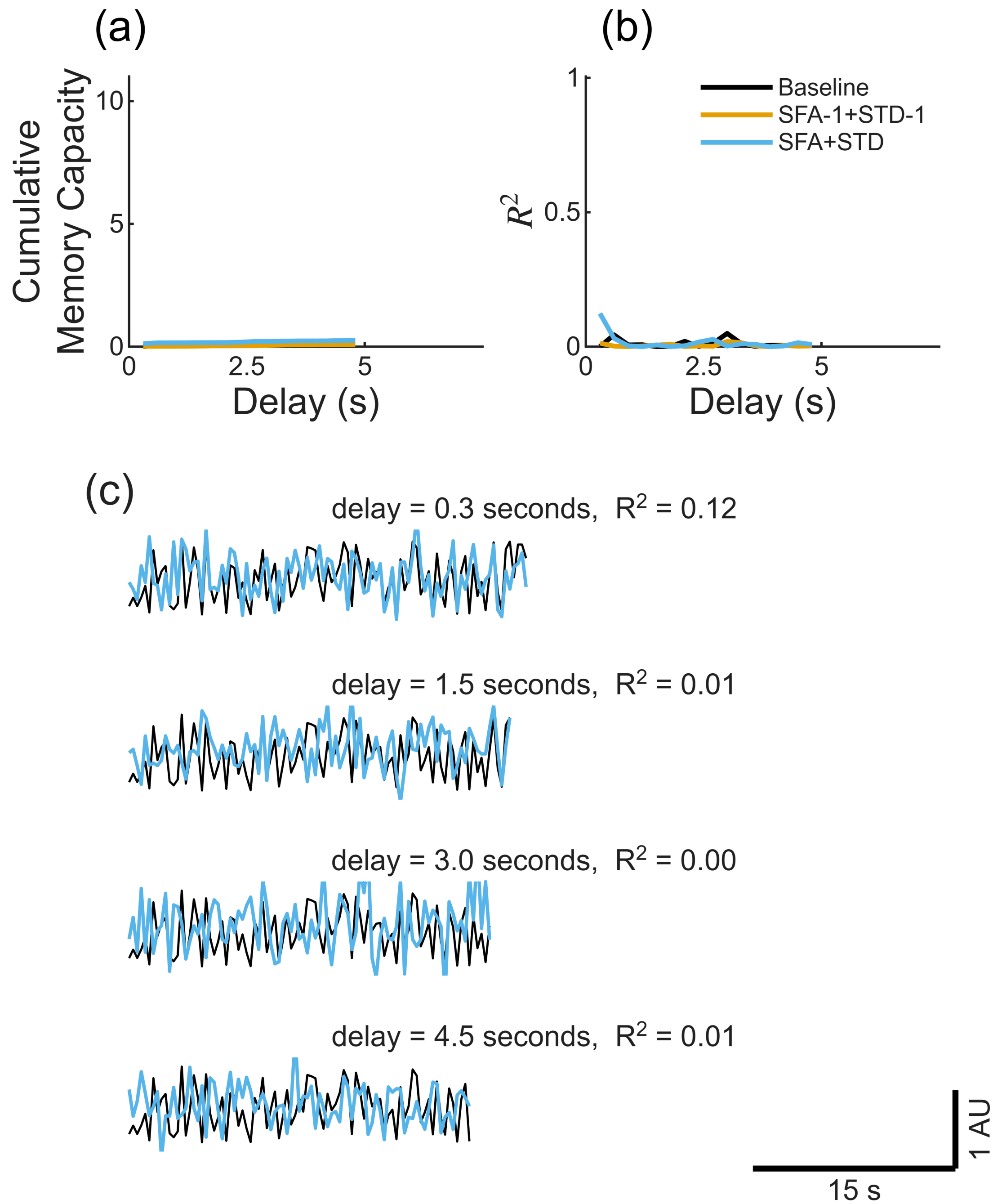
`fig_STD_steady_state/Fig_STD_steady_state.png`



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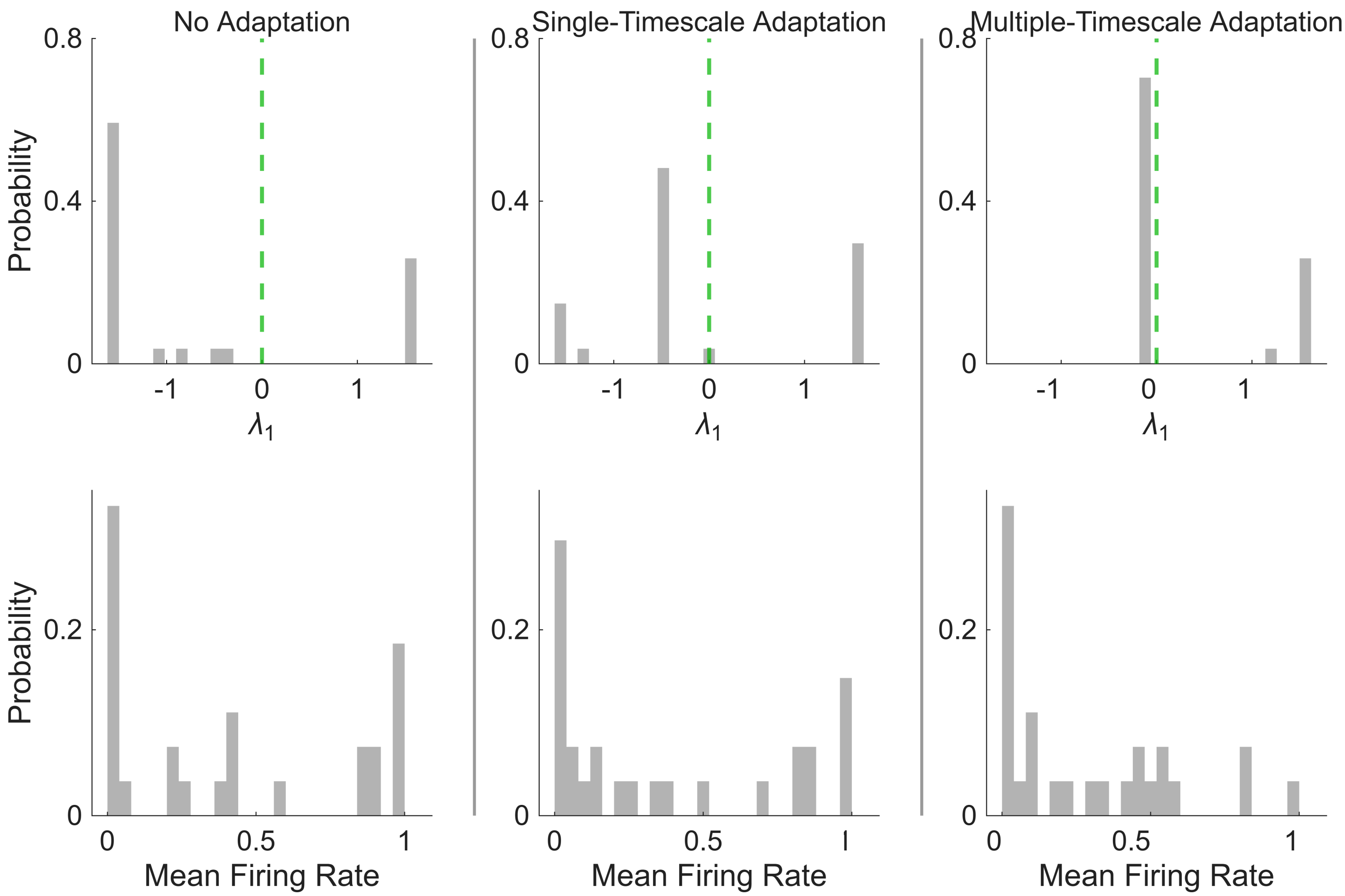

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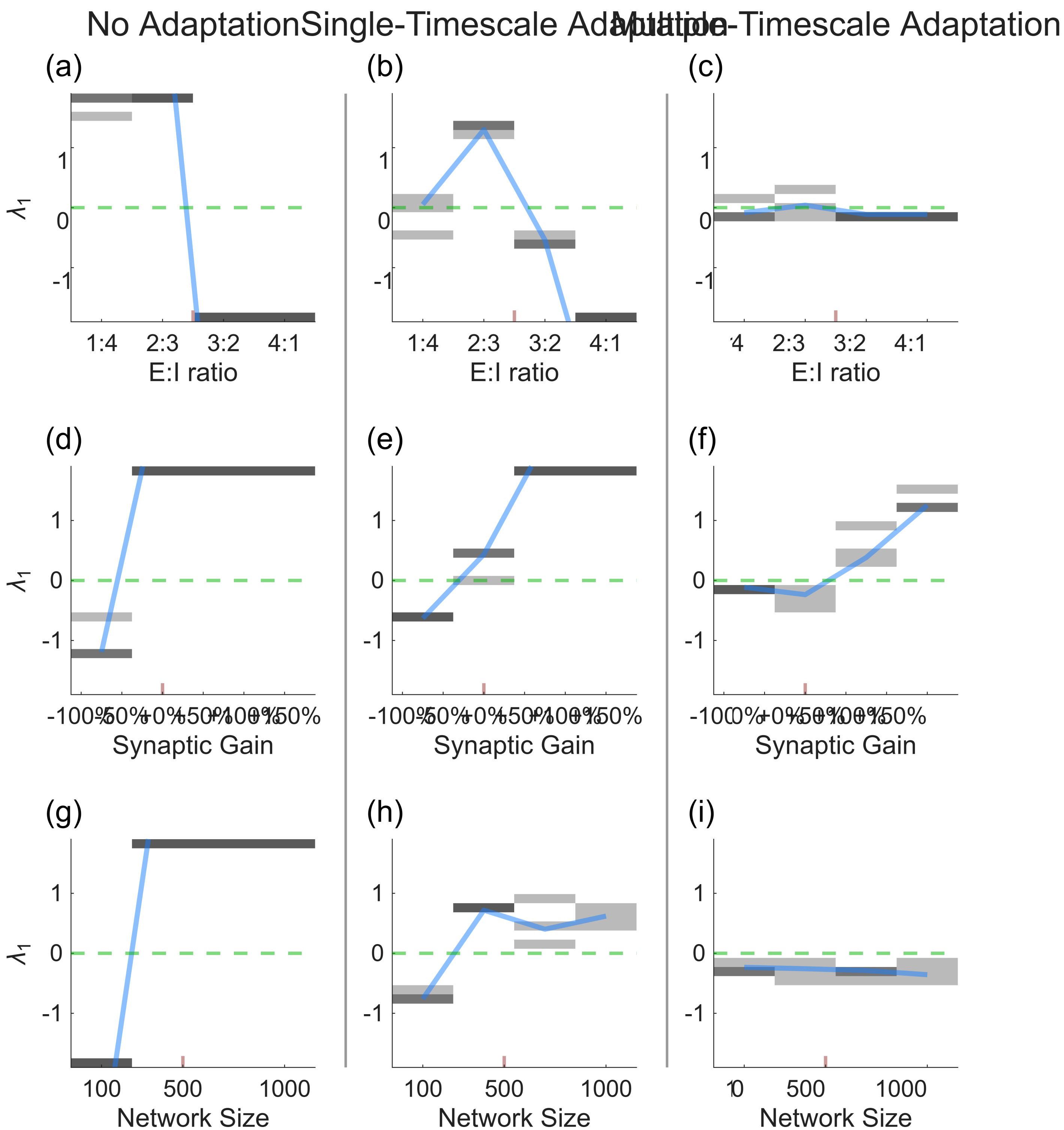
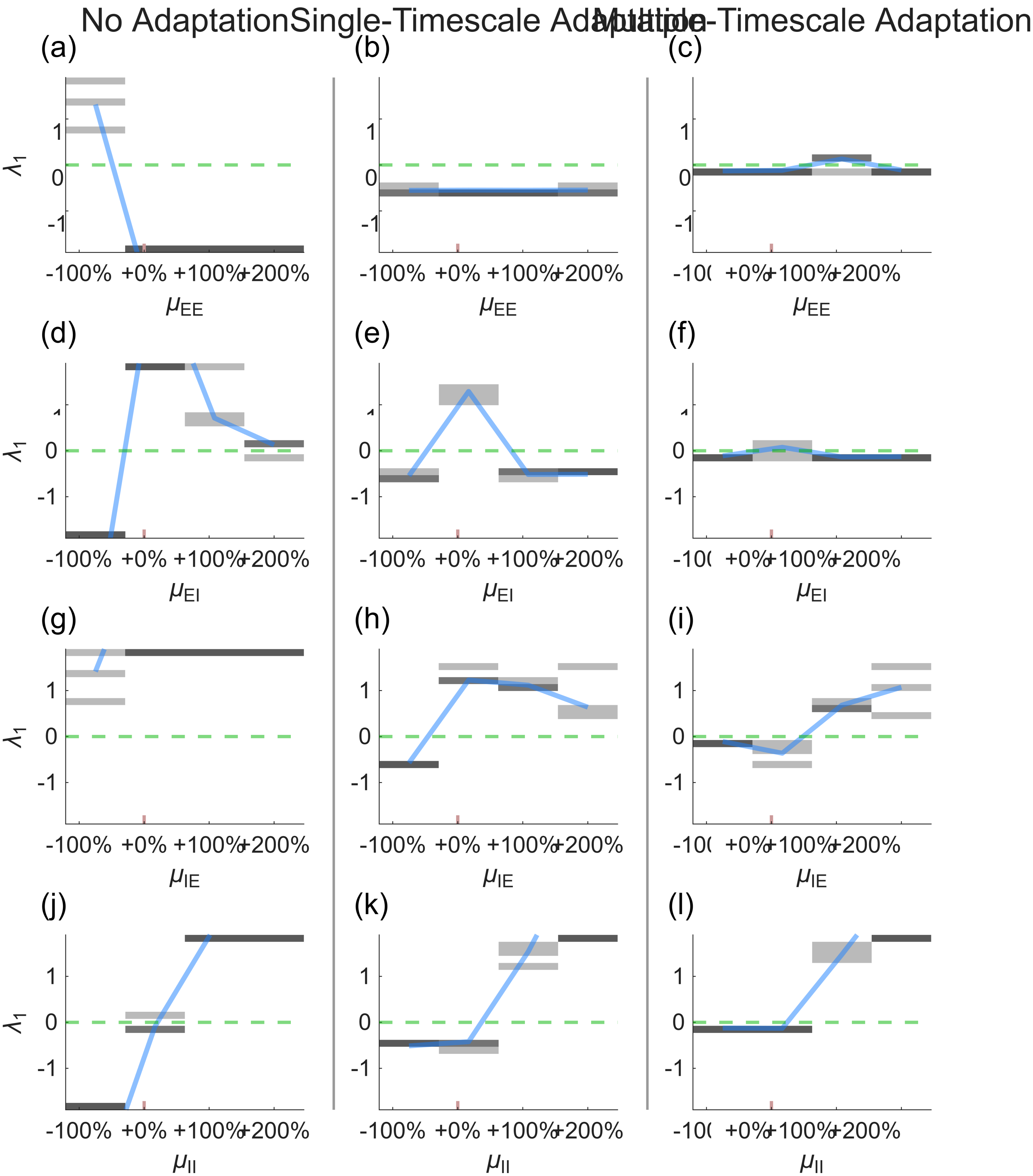


Figure 15: fig_sensitivity_analysis_allStd/Fig_Sensitivity_LLE_core.png



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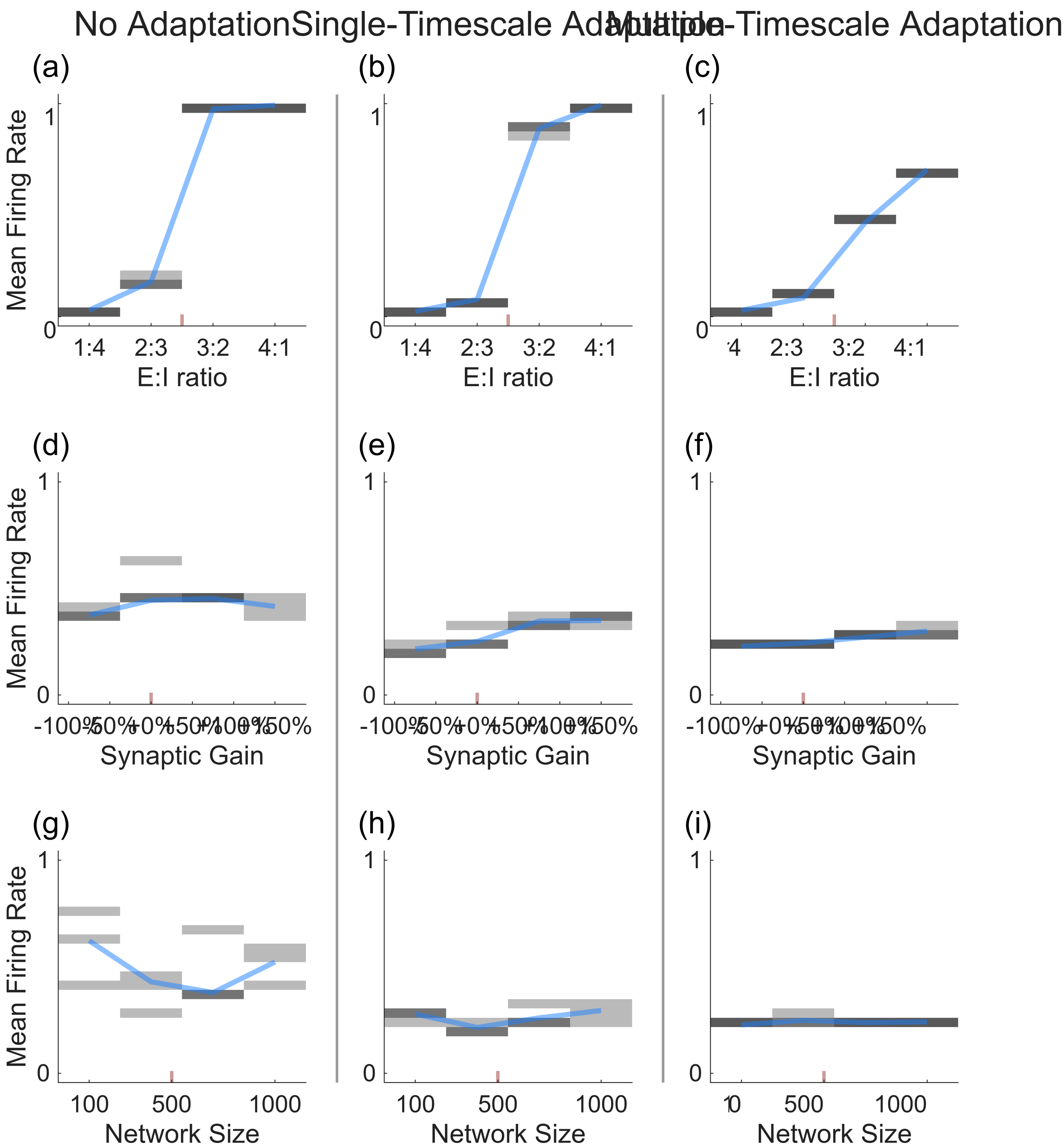
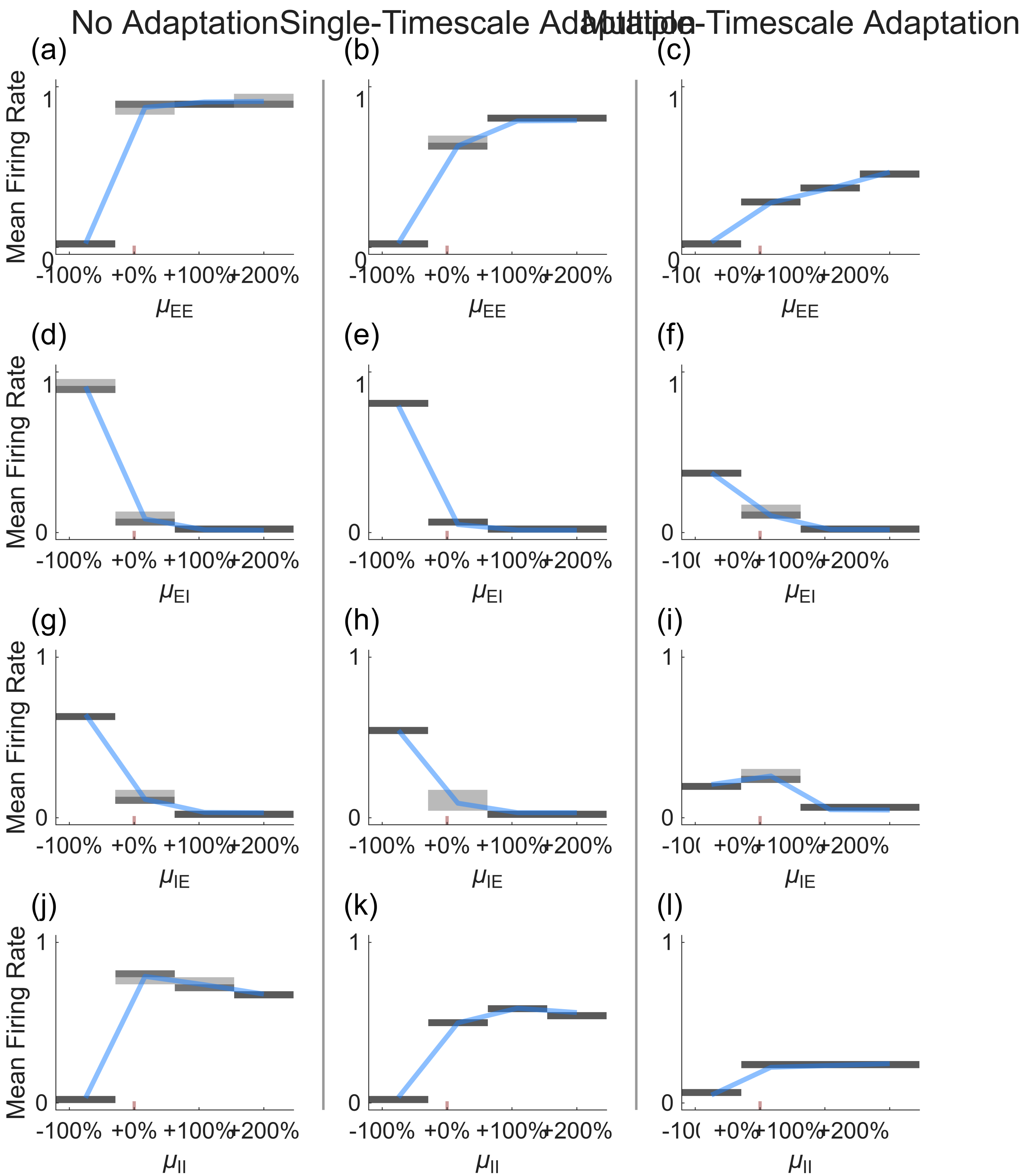


Figure 17: fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_core.png



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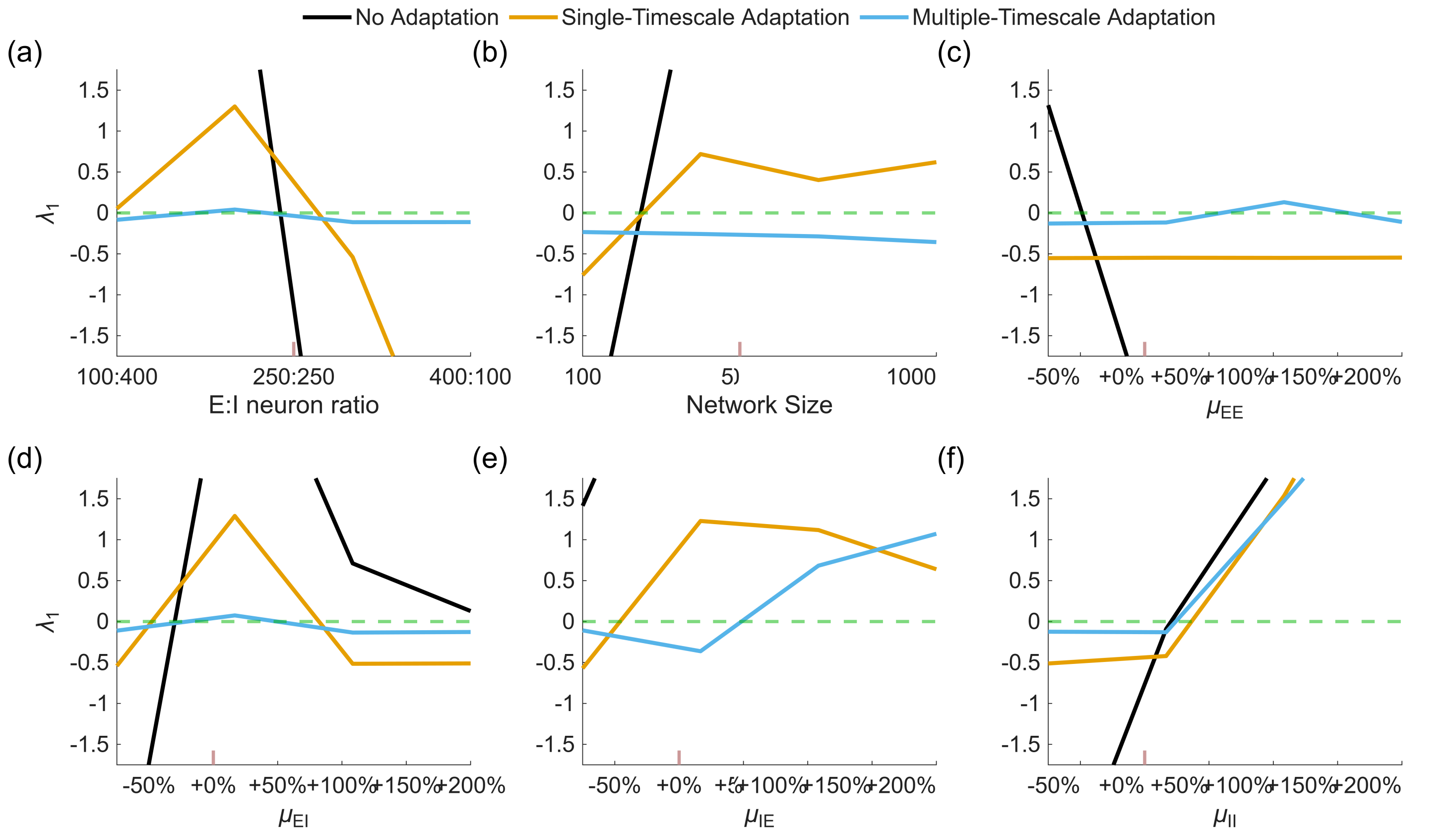

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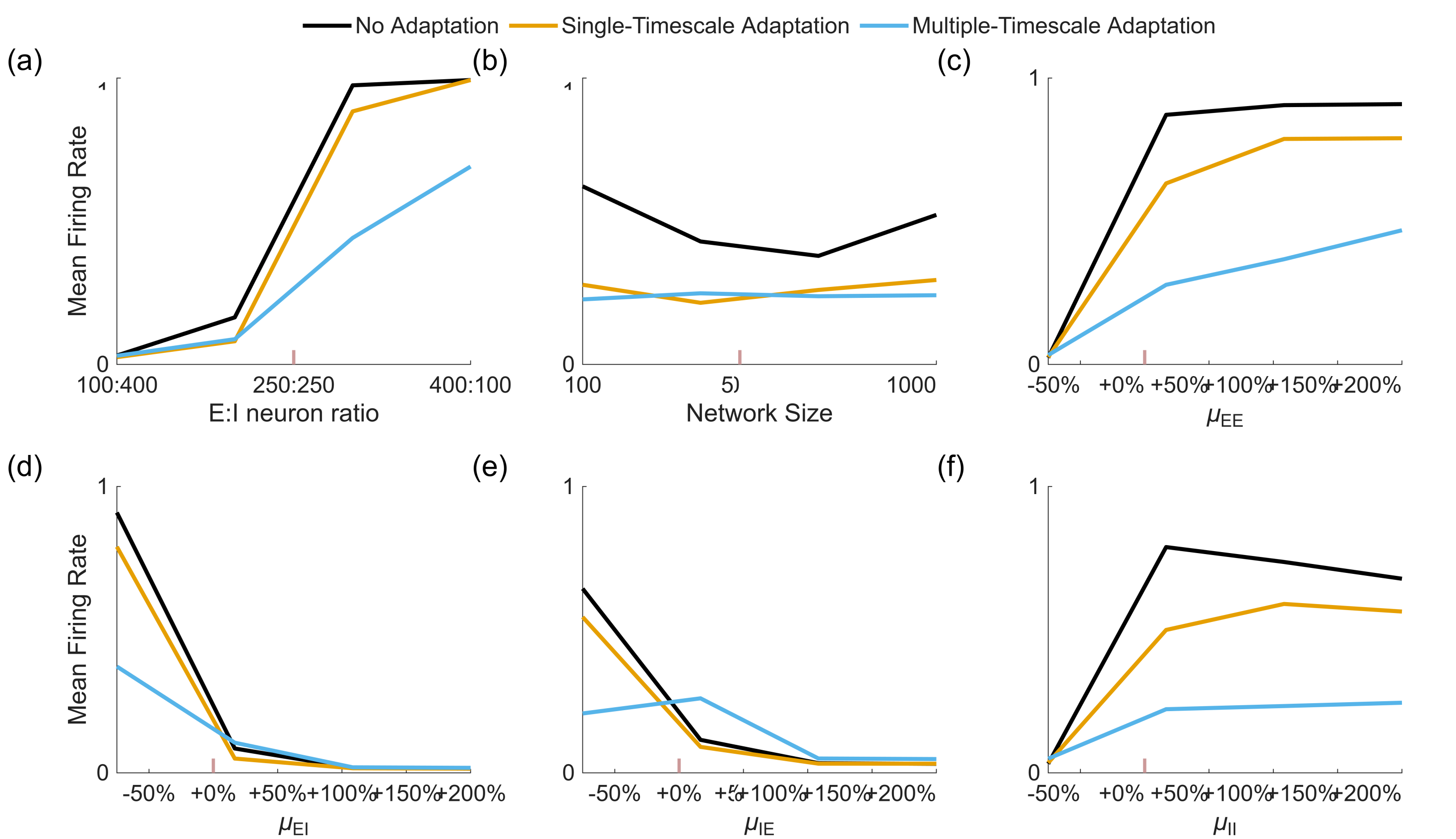

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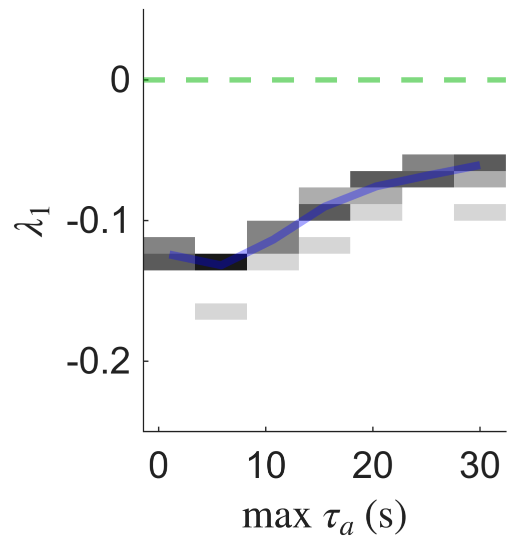

Figure 21: fig_sfa_EOC_allStd/Fig_SFA_EOC_allStd.png

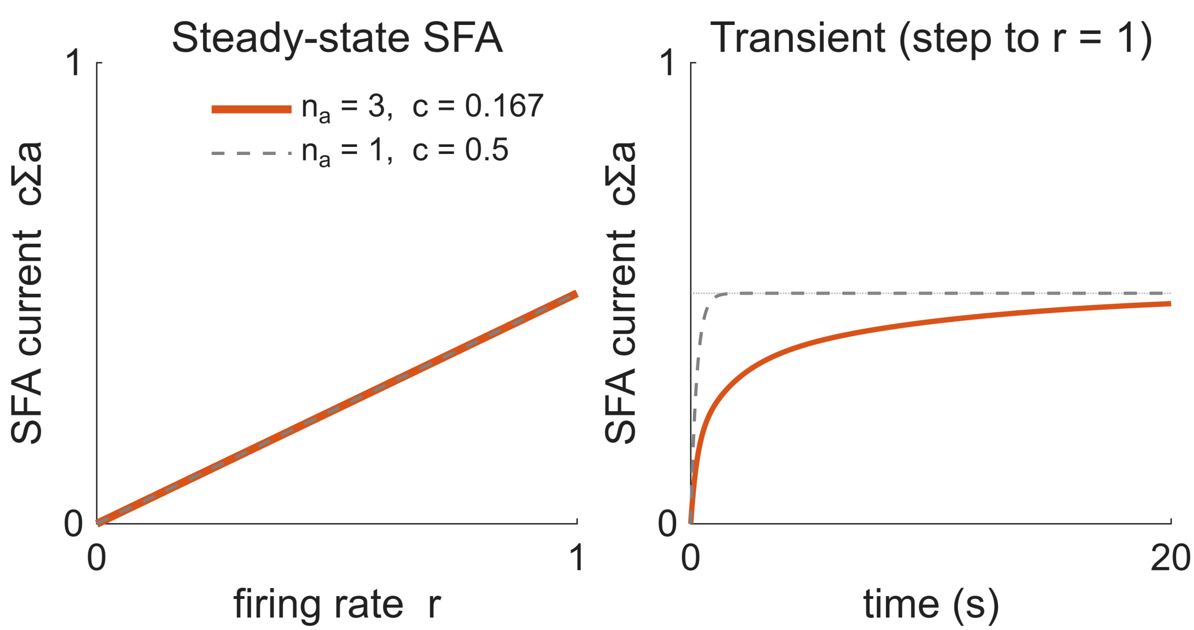

Figure 22: fig_SFA_steady_state/Fig_SFA_steady_state.png

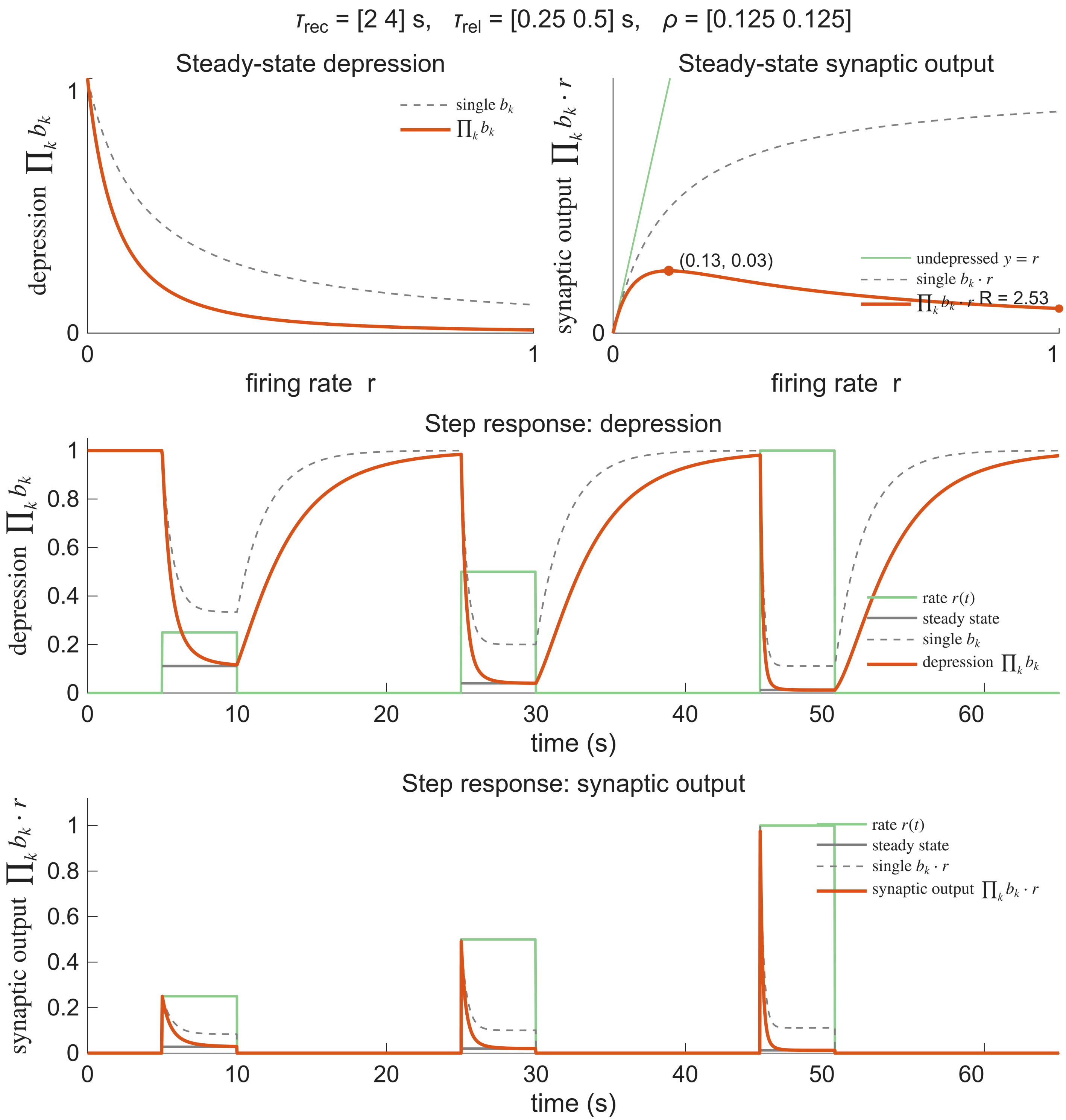

Figure 23: fig_STD_steady_state/Fig_STD_steady_state.png

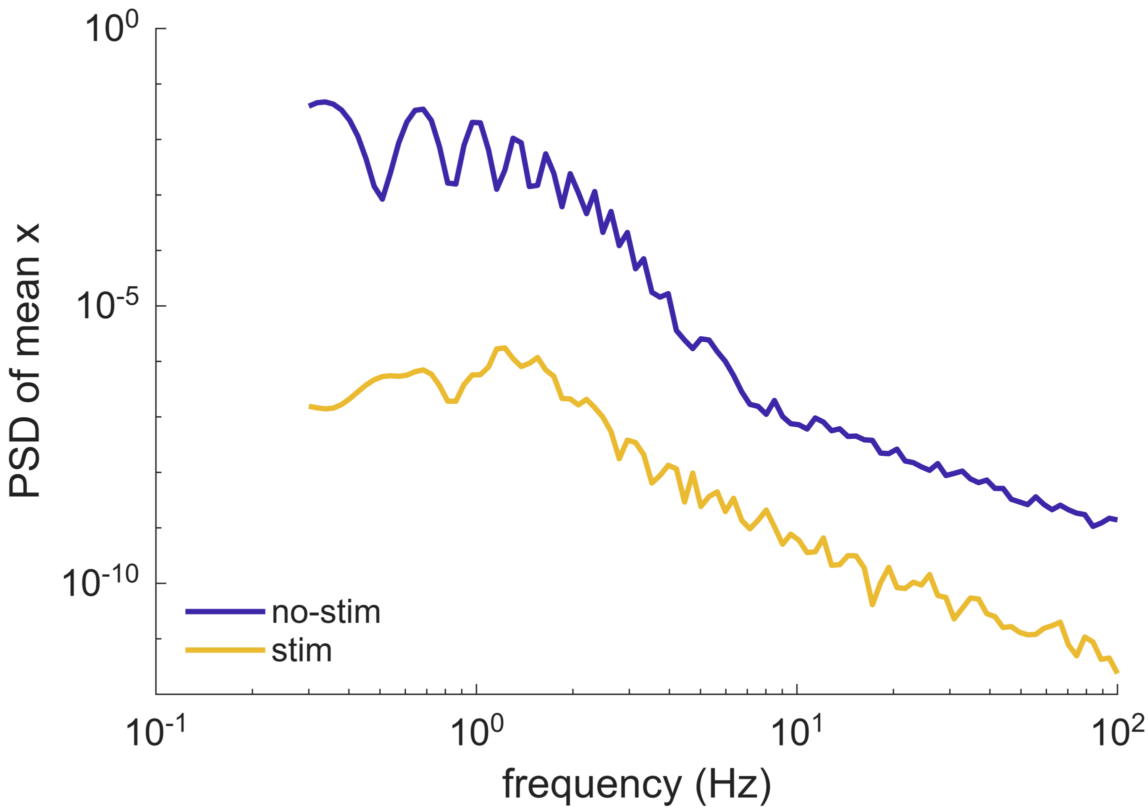

Figure 24: fig_stim_engages_adaptation/bursting_psd.png

fig_stim_engages_adaptation

`fig_stim_engages_adaptation/bursting_psd.png`

`fig_stim_engages_adaptation/bursting_synchrony.png`

`fig_stim_engages_adaptation/bursting_timeseries.png`

> Regenerate with `write_figure_report('C:\Users\m218089\Desktop\github_repos\FractionalReservoir\figs\single_`
> `figs/` is gitignored; this report references the images in place rather than copying them, so it is only ever as current as its generation stamp above.

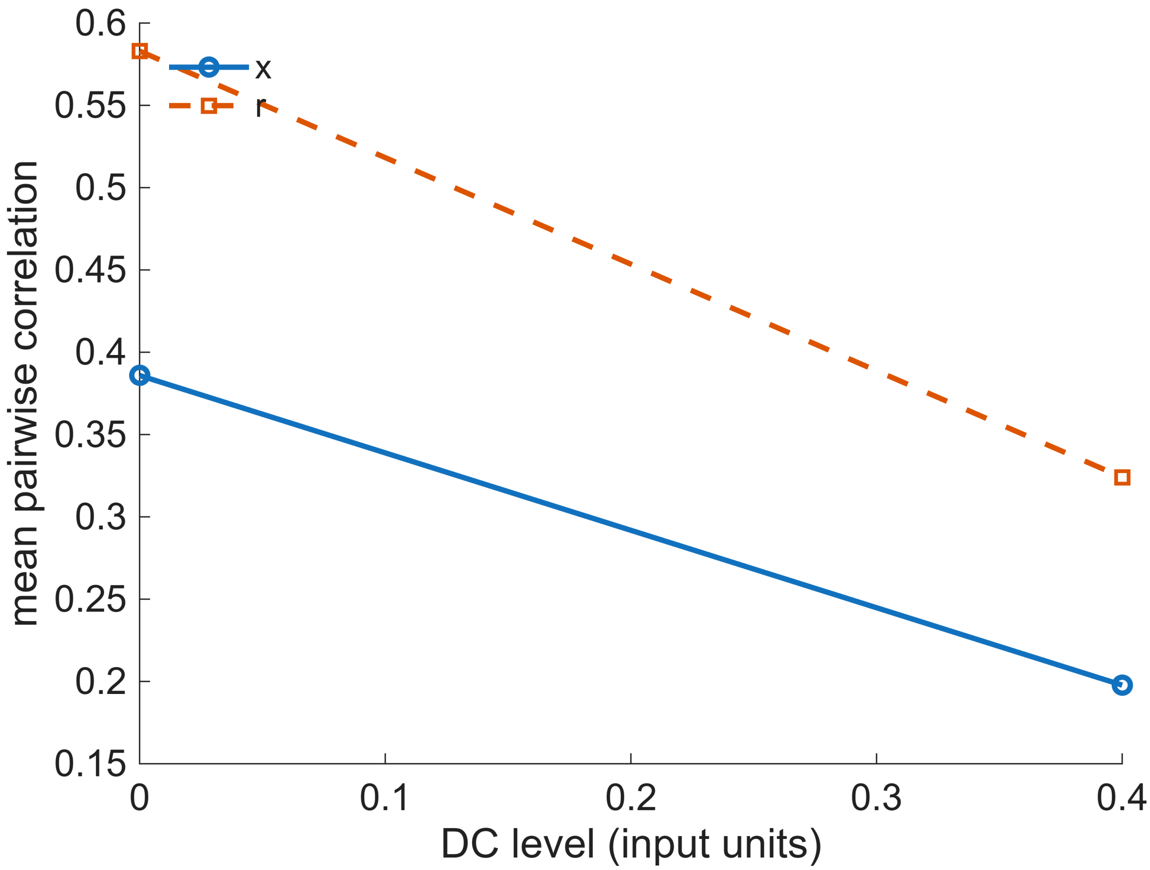

Figure 25: fig_stim_engages_adaptation/bursting_synchrony.png

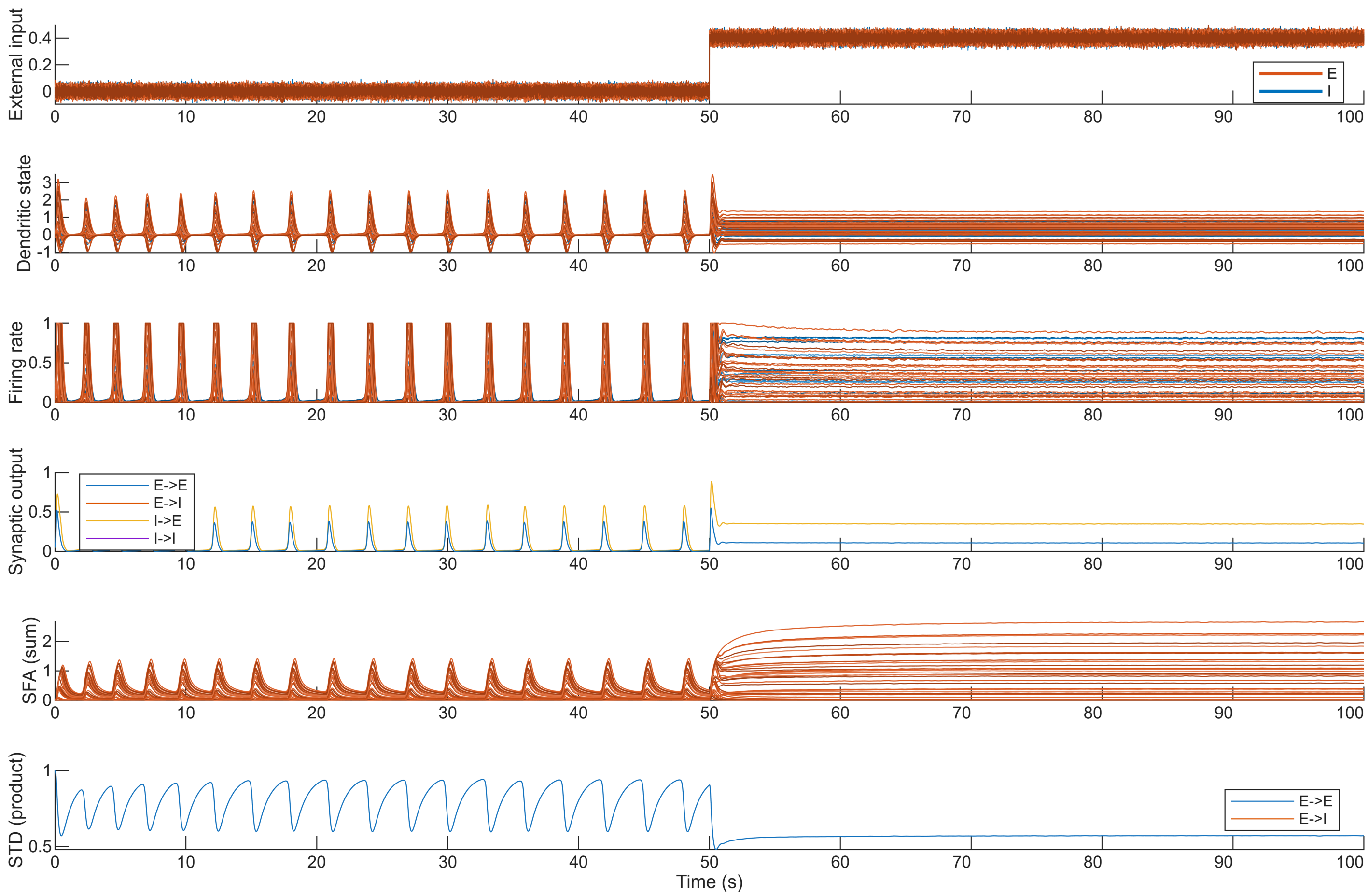

Figure 26: fig_stim_engages_adaptation/bursting_timeseries.png